

CEH Lab Report

Colton Schmid

Ferris State University

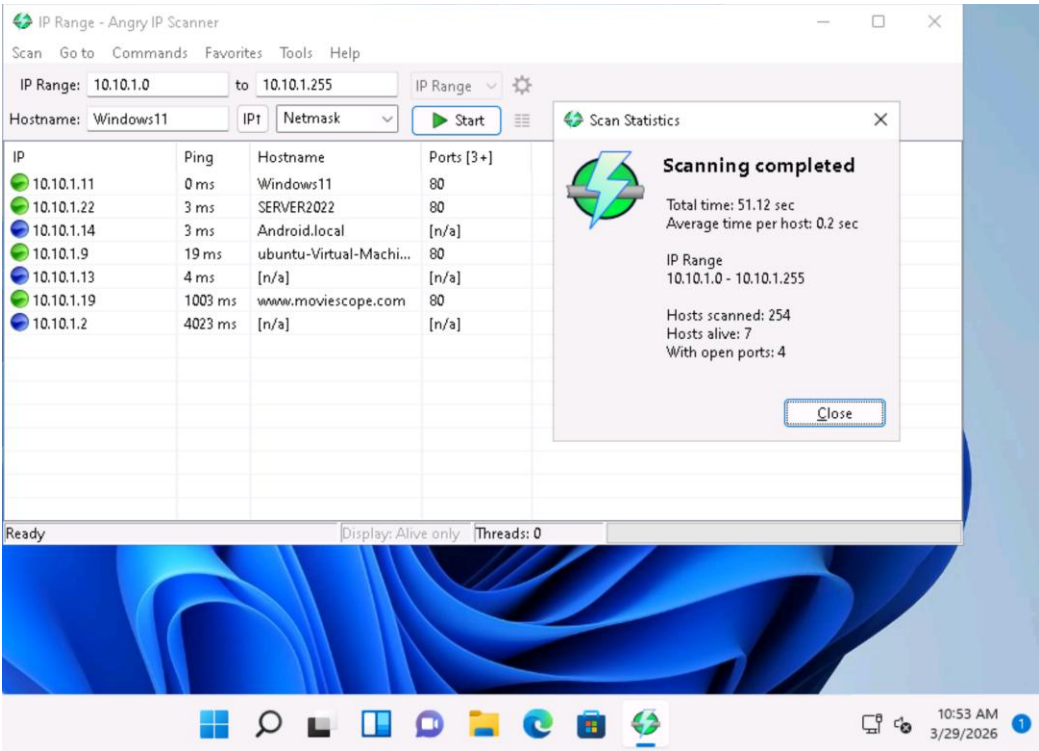
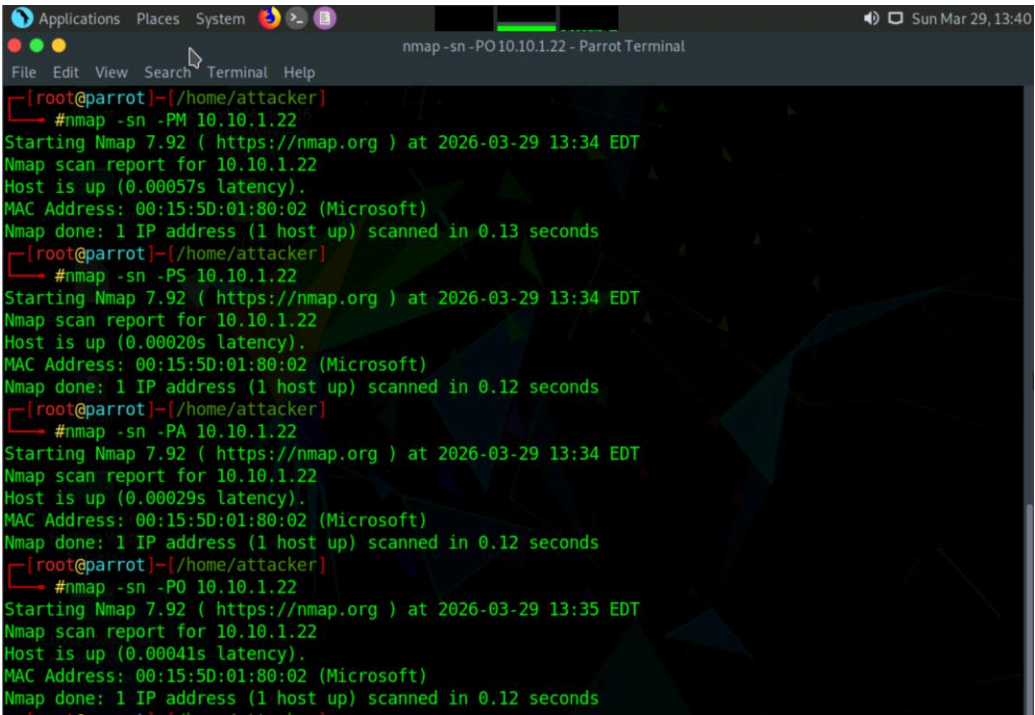
ISIN 312

Dr. Gerald Emerick

3/29/26

CEH Lab Report

Lab 1: Perform host discovery



Lab 2: Perform port and service discovery

Target: 10.10.1.* Profile: Scan Cancel

Command: nmap -A 10.10.1.*

NetScanTools Pro

Hosts Services Nmap Output Ports / Hosts Topology Host Details Scans

OS Host

- 10.10.1.2
- 10.10.1.9
- 10.10.1.11
- 10.10.1.13
- 10.10.1.14
- www.moviescope.c
- 10.10.1.22

Filter Hosts

10.10.1.22

Host Status

- State: up
- Open ports: 17
- Filtered ports: 0
- Closed ports: 983
- Scanned ports: 1000
- Up time: 4324
- Last boot: Sun Mar 29 10:23:15 2026

Addresses

- IPv4: 10.10.1.22
- IPv6: Not available
- MAC: 00:15:5D:01:80:02

Operating System

- Name: Microsoft Windows 10 1703
- Accuracy: 97%

Ports used

OS Classes

TCP Sequence

IP ID Sequence

TCP TS Sequence

1 item 1 item selected 92.9 MB State: Shared

11:42 AM 3/29/2026

```
Applications Places System hping3 -i 10.10.1.22 -p 80 -c 5 - Parrot Terminal
File Edit View Search Terminal Help

--- 10.10.1.22 hping statistic ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 7.5/7.7/7.8 ms
[root@parrot]~#
[root@parrot]~# hping3 --scan 0-100 -S 10.10.1.22
Scanning 10.10.1.22 (10.10.1.22), port 0-100
101 ports to scan, use -V to see all the replies
+-----+-----+-----+-----+
|port| serv name | flags | ttl | id | win | len |
+-----+-----+-----+-----+
53 domain   : .S..A... 128 33868 65392 44
80 http     : .S..A... 128 40780 65392 44
88 kerberos : .S..A... 128 43084 65392 44
All replies received. Done.
Not responding ports:
[root@parrot]~#
[root@parrot]~# hping3 -i 10.10.1.22 -p 80 -c 5
HPING 10.10.1.22 (eth0 10.10.1.22): icmp mode set, 28 headers + 0 data bytes
len=28 ip=10.10.1.22 ttl=128 id=19748 icmp_seq=0 rtt=7.8 ms
len=28 ip=10.10.1.22 ttl=128 id=19749 icmp_seq=1 rtt=7.8 ms
len=28 ip=10.10.1.22 ttl=128 id=19750 icmp_seq=2 rtt=7.7 ms
len=28 ip=10.10.1.22 ttl=128 id=19751 icmp_seq=3 rtt=11.6 ms
len=28 ip=10.10.1.22 ttl=128 id=19752 icmp_seq=4 rtt=3.5 ms
--- 10.10.1.22 hping statistic ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 3.5/7.7/11.6 ms
```

Lab 3: Perform OS discovery

```
Applications Places System nmap --script smb-os-discovery.nse 10.10.1.22 - Parrot Terminal
File Edit View Search Terminal Help
88/tcp open kerberos-sec
135/tcp open msrpc
139/tcp open netbios-ssn
389/tcp open ldap
445/tcp open microsoft-ds
464/tcp open kpasswd5
593/tcp open http-rpc-epmap
636/tcp open ldapssl
1801/tcp open msmq
2103/tcp open zephyr-clt
2105/tcp open eklogin
2107/tcp open msmq-mgmt
3268/tcp open globalcatLDAP
3269/tcp open globalcatLDAPssl
3389/tcp open ms-wbt-server
MAC Address: 00:15:5D:01:80:02 (Microsoft)

Host script results:
| smb-os-discovery:
|   OS: Windows Server 2022 Standard 20348 (Windows Server 2022 Standard 6.3)
|   Computer name: Server2022
|   NetBIOS computer name: SERVER2022\x00
|   Domain name: CEH.com
|   Forest name: CEH.com
|   FQDN: Server2022.CEH.com
|   System time: 2026-03-29T12:30:36-07:00
|_
Nmap done: 1 IP address (1 host up) scanned in 0.92 seconds
```

```
Applications Places System unicornscan 10.10.1.9 -lv - Parrot Terminal
File Edit View Search Terminal Help
TCP open ldap[ 389] from 10.10.1.22 ttl 128
TCP open microsoft-ds[ 445] from 10.10.1.22 ttl 128
TCP open ldaps[ 636] from 10.10.1.22 ttl 128
TCP open zephyr-clt[ 2103] from 10.10.1.22 ttl 128
TCP open ms-wbt-server[ 3389] from 10.10.1.22 ttl 128
[root@parrot]~#
#unicornscan 10.10.1.9 -lv
adding 10.10.1.9/32 mode 'TCPscan' ports `7,9,11,13,18,19,21-23,25,37,39,42,49,50,53,65,67-70,79-81,8
8,98,100,105-107,109-111,113,118,119,123,129,135,137-139,143,150,161-164,174,177-179,191,199-202,204,
206,209,210,213,220,345,346,347,369-372,389,406,407,422,443-445,487,500,512-514,517,518,520,525,533,5
38,548,554,563,587,610-612,631-634,636,642,653,655,657,666,706,750-752,765,779,808,873,901,923,941,94
6,992-995,1001,1023-1030,1080,1210,1214,1234,1241,1334,1349,1352,1423-1425,1433,1434,1524,1525,1645,1
646,1649,1701,1718,1719,1720,1723,1755,1812,1813,2048-2050,2101-2104,2140,2150,2233,2323,2345,2401,24
30,2431,2432,2433,2583,2628,2776,2777,2988,2989,3050,3130,3150,3232,3306,3389,3456,3493,3542-3545,363
2,3690,3801,4000,4400,4321,4567,4899,5002,5136-5139,5150,5151,5222,5269,5308,5354,5355,5422-5425,5432
,5503,5555,5556,5678,6000-6007,6346,6347,6543,6544,6789,6838,6666-6670,7000-7009,7028,7100,7983,8079-
8082,8088,8787,8879,9090,9101-9103,9325,9359,10000,10026,10027,10067,10080,10081,10167,10498,11201,15
345,17001-17003,18753,20011,20012,21554,22273,26274,27374,27444,27573,31335-31338,31787,31789,31790,3
1791,32668,32767-32780,33390,47262,49301,54320,54321,57341,58008,58009,58666,59211,60000,60006,61000,
61348,61466,61603,63485,63808,63809,64429,65000,65506,65530-65535' pps 300
using interface(s) eth0
scanning 1.00e+00 total hosts with 3.38e+02 total packets, should take a little longer than 8 Seconds
TCP open 10.10.1.9:80 ttl 64
TCP open 10.10.1.9:22 ttl 64
sender statistics 299.3 pps with 338 packets sent total
listener statistics 676 packets recieved 0 packets dropped and 0 interface drops
TCP open ssh[ 22] from 10.10.1.9 ttl 64
TCP open http[ 80] from 10.10.1.9 ttl 64
```

Lab 4: Scan beyond IDS and Firewall

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
39598	492.520733	10.10.1.13	10.10.1.11	TCP	74	40088 → 139 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
39599	492.520853	10.10.1.11	10.10.1.13	TCP	74	139 → 40088 [SYN, ACK] Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=...
39600	492.520936	10.10.1.13	10.10.1.11	TCP	74	45886 → 8888 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
39601	492.521005	10.10.1.13	10.10.1.11	TCP	66	40088 → 139 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=7014886 TS...
39602	492.521045	10.10.1.13	10.10.1.11	TCP	66	40088 → 139 [RST, ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=70148...
39603	492.521337	10.10.1.13	10.10.1.11	TCP	74	53080 → 110 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
39604	492.521445	10.10.1.13	10.10.1.11	TCP	74	45136 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS...
39605	492.521508	10.10.1.11	10.10.1.13	TCP	74	80 → 45136 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=...
39606	492.521686	10.10.1.13	10.10.1.11	TCP	66	45136 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=7014886 TS...
39607	492.521688	10.10.1.13	10.10.1.11	TCP	74	47584 → 5900 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
39608	492.521720	10.10.1.13	10.10.1.11	TCP	66	45136 → 80 [RST, ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=70148...
39609	492.521884	10.10.1.13	10.10.1.11	TCP	74	60836 → 113 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
39610	492.522151	10.10.1.13	10.10.1.11	TCP	74	53358 → 143 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
39611	492.522373	10.10.1.13	10.10.1.11	TCP	74	60494 → 23 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS...
39612	492.522645	10.10.1.13	10.10.1.11	TCP	74	59170 → 1025 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
39613	492.523027	10.10.1.13	10.10.1.11	TCP	74	59078 → 3306 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
39614	492.523131	10.10.1.13	10.10.1.11	TCP	74	43860 → 3389 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
39615	492.523188	10.10.1.11	10.10.1.13	TCP	74	3389 → 43860 [SYN, ACK] Seq=0 Ack=1 Win=64000 Len=0 MSS=1460 S...
39616	492.523369	10.10.1.13	10.10.1.11	TCP	66	43860 → 3389 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=7014888 TS...
39617	492.523391	10.10.1.13	10.10.1.11	TCP	66	43860 → 3389 [RST, ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=7014...
39618	492.523675	10.10.1.13	10.10.1.11	TCP	74	53378 → 1723 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T...
39619	492.523975	10.10.1.13	10.10.1.11	TCP	74	59152 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TS...
39620	493.535773	10.10.1.13	10.10.1.11	TCP	74	[TCP Retransmission] [TCP Port numbers reused] 59170 → 1025 [S...
39621	493.535911	10.10.1.13	10.10.1.11	TCP	74	[TCP Retransmission] [TCP Port numbers reused] 59152 → 22 [SYN...
39622	493.536294	10.10.1.13	10.10.1.11	TCP	74	[TCP Retransmission] [TCP Port numbers reused] 60494 → 23 [SYN...
39623	493.536559	10.10.1.13	10.10.1.11	TCP	74	[TCP Retransmission] [TCP Port numbers reused] 53358 → 143 [SY...
39624	493.536713	10.10.1.13	10.10.1.11	TCP	74	[TCP Retransmission] [TCP Port numbers reused] 53378 → 1723 [S...
39625	493.537112	10.10.1.13	10.10.1.11	TCP	74	[TCP Retransmission] [TCP Port numbers reused] 60836 → 113 [SY...
39626	493.537337	10.10.1.13	10.10.1.11	TCP	74	[TCP Retransmission] [TCP Port numbers reused] 59078 → 3306 [S...

> Frame 1: 418 bytes on wire (3344 bits), 418 bytes captured (3344 bits) on interface \Device\NPF_{5A9B3588-F693-4023-B906-DCC29AD0B114}, id 0

> Ethernet II, Src: MS-NLB-PhysServer-21_5d:42:8d:52 (02:15:5d:42:8d:52), Dst: IP4mcast fb (01:00:5e:00:00:fb)

0000 01 00 5e 00 00 fb 02 15 5d 42 8d 52 08 00 45 00]B.Q.E

0010 01 94 60 81 40 00 ff 11 2d c4 0a 0a 01 0e 0e 00@.....

Ethernet: <live capture in progress>

Packets: 41677 · Displayed: 41677 (100.0%) Profile: Default

12:59 PM 3/29/2026

No.	Time	Source	Destination	Protocol	Length	Info
2402.	225.324396	10.10.1.13	10.10.1.11	TCP	54	55042 → 0 [None] Seq=4266095696 Win=512 Len=0
2402.	225.324396	10.10.1.13	10.10.1.11	TCP	54	55053 → 0 [None] Seq=4277732635 Win=512 Len=0
2402.	225.324396	10.10.1.13	10.10.1.11	TCP	54	55060 → 0 [None] Seq=268176735 Win=512 Len=0
2402.	225.324396	10.10.1.13	10.10.1.11	TCP	54	55063 → 0 [None] Seq=4246382477 Win=512 Len=0
2402.	225.324421	10.10.1.13	10.10.1.11	TCP	54	55070 → 0 [None] Seq=3957616977 Win=512 Len=0
2402.	225.324430	10.10.1.13	10.10.1.11	TCP	54	55072 → 0 [None] Seq=3680518764 Win=512 Len=0
2402.	225.324430	10.10.1.13	10.10.1.11	TCP	54	55075 → 0 [None] Seq=223366424 Win=512 Len=0
2402.	225.324443	10.10.1.13	10.10.1.11	TCP	54	55012 → 0 [None] Seq=3917375541 Win=512 Len=0
2402.	225.324443	10.10.1.13	10.10.1.11	TCP	54	55021 → 0 [None] Seq=1425405087 Win=512 Len=0
2402.	225.324443	10.10.1.13	10.10.1.11	TCP	54	55047 → 0 [None] Seq=3831975215 Win=512 Len=0
2402.	225.324443	10.10.1.13	10.10.1.11	TCP	54	55067 → 0 [None] Seq=3156397163 Win=512 Len=0
2402.	225.324446	10.10.1.13	10.10.1.11	TCP	54	54966 → 0 [None] Seq=4021585701 Win=512 Len=0
2402.	225.324446	10.10.1.13	10.10.1.11	TCP	54	54978 → 0 [None] Seq=664356603 Win=512 Len=0
2402.	225.324446	10.10.1.13	10.10.1.11	TCP	54	54987 → 0 [None] Seq=1443500904 Win=512 Len=0
2402.	225.324446	10.10.1.13	10.10.1.11	TCP	54	54996 → 0 [None] Seq=74393405 Win=512 Len=0
2402.	225.324446	10.10.1.13	10.10.1.11	TCP	54	55003 → 0 [None] Seq=3851331240 Win=512 Len=0
2402.	225.324446	10.10.1.13	10.10.1.11	TCP	54	55017 → 0 [None] Seq=1120600641 Win=512 Len=0
2402.	225.324446	10.10.1.13	10.10.1.11	TCP	54	55018 → 0 [None] Seq=134656812 Win=512 Len=0
2402.	225.324446	10.10.1.13	10.10.1.11	TCP	54	55020 → 0 [None] Seq=2259042876 Win=512 Len=0
2402.	225.324446	10.10.1.13	10.10.1.11	TCP	54	55033 → 0 [None] Seq=3937067536 Win=512 Len=0
2402.	225.324461	10.10.1.13	10.10.1.11	TCP	54	55077 → 0 [None] Seq=4100022005 Win=512 Len=0

> Frame 2402575: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{5A9B3588-F693-4023-B906-DCC29AD0B114}, id 0

> Ethernet II, Src: MS-NLB-PhysServer-21_5d:42:8d:51 (02:15:5d:42:8d:51), Dst: Microsoft_01:80:00 (00:15:5d:01:80:00)

> Internet Protocol Version 4, Src: 10.10.1.13, Dst: 10.10.1.11

> Transmission Control Protocol, Src Port: 54996, Dst Port: 0, Seq: 74393405, Len: 0

0000 00 15 5d 01 80 00 02 15 5d 42 8d 51 08 00 45 00]B.Q.E

0010 00 28 2f c8 00 00 04 06 34 dd 0a 0a 01 0d 0a 0a(.....4.....

wreshark_EthernetX23QM3.pcapng

Packets: 2402581 · Displayed: 2402581 (100.0%) · Dropped: 211602 (8.8%) Profile: Default

1:11 PM 3/29/2026

Lab 5: Perform network scanning using various scanning tools

```
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
RHOSTS => file:/tmp/msf-db-rhosts-20260329-3790-1hgf6u
msf6 auxiliary(scanner/portscan/tcp) > set RHOSTS 10.10.1.22
RHOSTS => 10.10.1.22
msf6 auxiliary(scanner/portscan/tcp) > RUN
[-] Unknown command: RUN
msf6 auxiliary(scanner/portscan/tcp) > run

[+] 10.10.1.22: - 10.10.1.22:53 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:80 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:88 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:135 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:139 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:389 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:445 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:464 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:593 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:636 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:1801 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:2103 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:2105 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:2107 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:3268 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:3269 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:3389 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:5985 - TCP OPEN
[+] 10.10.1.22: - 10.10.1.22:9389 - TCP OPEN
[*] 10.10.1.22: - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/portscan/tcp) >
```

```
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
msf6 auxiliary(scanner/smb/smb_version) > set RHOSTS 10.10.1.5-23
RHOSTS => 10.10.1.5-23
msf6 auxiliary(scanner/smb/smb_version) > set THREADS 11
THREADS => 11
msf6 auxiliary(scanner/smb/smb_version) > run

[*] 10.10.1.11:445 - SMB Detected (versions:1, 2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:LZNT1, Pattern V1) (encryption capabilities:AES-256-GCM) (signatures:optional) (guid:{659890e9-e776-4110-903f-7e28a80b87fa}) (authentication domain:WINDOWS11)
[+] 10.10.1.11:445 - Host is running Windows 10 Enterprise (build:22000) (name:WINDOWS11) (workgroup:WORKGROUP)
[*] 10.10.1.5-23: - Scanned 4 of 19 hosts (21% complete)
[*] 10.10.1.19:445 - SMB Detected (versions:2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:) (encryption capabilities:AES-128-GCM) (signatures:optional) (guid:{4d8e08ce-f381-451d-a96c-15ec7de62a8b}) (authentication domain:SERVER2019)
[*] 10.10.1.5-23: - Scanned 5 of 19 hosts (26% complete)
[*] 10.10.1.5-23: - Scanned 8 of 19 hosts (42% complete)
[*] 10.10.1.5-23: - Scanned 9 of 19 hosts (47% complete)
[*] 10.10.1.5-23: - Scanned 12 of 19 hosts (63% complete)
[*] 10.10.1.22:445 - SMB Detected (versions:1, 2, 3) (preferred dialect:SMB 3.1.1) (compression capabilities:LZNT1, Pattern V1) (encryption capabilities:AES-256-GCM) (signatures:required) (guid:{62cb239a-9011-4728-88c3-21e8212b0eb9}) (authentication domain:CEH)
[*] 10.10.1.22:445 - Host could not be identified: Windows Server 2022 Standard 20348 (Windows Server 2022 Standard 6.3)
[*] 10.10.1.5-23: - Scanned 16 of 19 hosts (84% complete)
[*] 10.10.1.5-23: - Scanned 17 of 19 hosts (89% complete)
[*] 10.10.1.5-23: - Scanned 17 of 19 hosts (89% complete)
[*] 10.10.1.5-23: - Scanned 19 of 19 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/smb/smb_version) >
```

Critical Analysis, Ethical Considerations, and Reflection

Throughout these five labs I learned multiple skills and tools that can help me when scanning a network. This lab started off with discovering hosts on the network. For this we used both Nmap and a scanning software called Angry IP Scanner. This made me a lot more comfortable with the Nmap command and the syntax to capture specific information. This lab taught me a lot about Nmap. To explain, I did not know all the different uses that Nmap had, such as OS discovery, and this was very cool to me. Additionally, this lab emphasized the importance of a properly configured firewall. These labs did this because it had us turn on and off the firewall demonstrated ways to work around the rules that are in place. To explain, we used packet fragmentation and port manipulation to work around the firewall.

When it comes to ethical considerations for these labs, using these tools can be used for many different ethical reasons. One being that these techniques and tools could be used to test how strong an organization's firewall configuration is. Additionally, these could also be used to assess an automation detection system.

This assignment took me around 3 hours to complete; I was in between different things while doing the labs. The main thing that surprised me in these labs was the versatility of the Nmap command. This surprised me because I was not expecting it to be able to do everything that we completed in this lab. I thought I was familiar with Nmap, well I was with its basic commands, but I want to fully look at the help page for Nmap and fully understand its capabilities. Another area that I am more interested in is the `hping3` command. This interests me, similar to the Nmap command, because of its versatility.